CONTEXTFLUX: CONDITIONING A FROZEN FLOW-MATCHING TRANSFORMER WITH BLUR-ROBUST V-JEPA CONTEXT FOR MOTION DEBLURRING

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ABSTRACT

Diffusion-based image restoration methods condition large text-to-image backbones on captions extracted from the degraded input, but captioning a blurred image is unreliable and discards spatial detail. We ask whether the patch representations of a self-supervised video encoder, V-JEPA, can replace text conditioning entirely. We first analyze V-JEPA features under motion blur and find that they remain strongly correlated with the features of the sharp image, making them a natural blur-robust conditioning signal. Building on this, we propose *ContextFlux*: a lightweight dual-head cross-attention projector maps frozen V-JEPA patch tokens into (i) the text-conditioning space and (ii) the packed VAE-latent space of a completely frozen FLUX.2 flow-matching transformer. The projector is first aligned to caption embeddings and clean-image latents, then fine-tuned with the flow-matching objective while the 4B-parameter backbone, the VAE, and the encoder all stay frozen — the projector is the only trained component. On a synthetic motion-blur benchmark, we compare V-JEPA context against DINOv2, CLIP, and VL-JEPA conditioning, a LoRA-tuned image-conditioned control, and regression baselines, and ablate feature depth and feature source. [Finalize abstract claims once the unified evaluation completes (plan step 5).]

***Index Terms*—** Image deblurring, flow matching, self-supervised representations, JEPA, diffusion models

1. INTRODUCTION

Recovering a sharp image from a motion-blurred observation is a long-standing inverse problem. Regression networks such as Restormer [1] and NAFNet [2] achieve strong distortion metrics but over-smooth fine texture. Generative restoration instead exploits the prior of a large pre-trained text-to-image model [3, 4, 5], and recent flow-matching transformers [6, 7, 8] make this prior stronger and cheaper to sample.

To steer such a backbone toward the latent clean image, current pipelines extract *semantic* conditioning from the degraded input — typically a caption or tag set produced by a vision–language model, following the caption-free-restoration line of LucidFlux [9][verify LucidFlux citation details]. This is doubly unsatisfying for deblurring: a VLM applied to a blurred image produces unreliable text, and even a perfect caption is a severe information bottleneck — it carries no spatial layout.

We instead condition on the degraded image's *self-supervised patch representations*. V-JEPA [10, 11][verify V-JEPA 2 citation] learns by predicting the embeddings of masked regions in latent space, an objective that encourages abstraction over low-level appearance. We hypothesize — and verify empirically in Sec. 4 — that this makes V-JEPA patch tokens substantially *blur-robust*: features of a blurred image stay close to those of its sharp counterpart. Such tokens are then a conditioning signal that (a) survives the degradation, (b) retains spatial structure that captions lose, and (c) is available without any VLM at test time.

Our system, **ContextFlux**, keeps the entire FLUX.2 Klein backbone (transformer, VAE, text pathway) and the V-JEPA encoder frozen. A single dual-head cross-attention projector (∼—M parameters[compute from checkpoint]) maps V-JEPA tokens into the backbone's text-token stream and, through a second head, into its packed VAE-latent token stream. Notably, in our main configuration the low-quality (LQ) image latents are *not* fed to the transformer at all: every bit of information about the input image reaches the backbone through projected V-JEPA tokens. Deblurring quality therefore directly measures how much restoration-relevant content the self-supervised representation carries.

Contributions. (1) An empirical analysis of V-JEPA representations under motion blur (spectrum, per-component invariance, and GT-vs-blur token similarity) compared against DINOv2 and CLIP; (2) ContextFlux, a two-stage recipe — caption/latent alignment, then flow-matching fine-tuning — that conditions a fully frozen flow transformer on projected V-JEPA tokens only; (3) a controlled comparison across conditioning encoders (V-JEPA, DINOv2, CLIP, VL-JEPA), feature depth, and feature source (degraded vs. oracle-sharp), against regression and LoRA-tuned diffusion baselines.

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2. RELATED WORK

Deblurring. Multi-scale CNNs [16] evolved into restoration transformers [1, 17] and simplified baselines [2]. Diffusion-prior methods [4, 3, 5] trade distortion for perceptual quality by sampling from a pre-trained generative model conditioned on the degraded input.

Flow matching. Rectified-flow transformers [6, 18, 7] learn a velocity field transporting noise to data; FLUX [8] scales this recipe. We build on the distilled FLUX.2 Klein model, which admits guidance-free few-step sampling.

Conditioning restoration without captions. Lucid-Flux [9] conditions a Flux backbone for universal restoration without text captions; we go further by removing image-latent conditioning as well, isolating the representation itself as the sole carrier of image content, and by systematically comparing self-supervised encoders. Joint-embedding predictive architectures [10, 11] have not previously been used as the conditioning interface of a generative restorer; DINOv2 [12] and CLIP [13] serve as our contrastive/distillation-based comparisons.

3. METHOD

3.1. Preliminaries

FLUX.2 Klein is a flow-matching transformer v_θ operating on packed VAE latents. Training minimizes

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1}\left[w(t)\left\|v_\theta(\mathbf{x}_t,t,\mathbf{c}) - (\mathbf{x}_0-\mathbf{x}_1)\right\|^2\right], \quad (1)$$

with $\mathbf{x}_t = (1-\sigma_t)\mathbf{x}_1 + \sigma_t\mathbf{x}_0$ and a discrete shifted schedule (shift 3.1582 in training; resolution-dependent dynamic shifting at sampling). The conditioning $\mathbf{c}$ enters through a token stream processed jointly with the latent tokens; in text-to-image use, $\mathbf{c}$ holds text-encoder embeddings.

3.2. ContextFlux conditioning

Given LQ input $\mathbf{y}$, an optional frozen SwinIR [15] pre-filter produces $\tilde{\mathbf{y}}$, from which the frozen V-JEPA 2.1 encoder (ViT-L, 512^2 input) extracts patch tokens $\mathbf{F} \in \mathbb{R}^{N_p\times d_J}$ (N_p=1024, d_J=1024; concatenating the last N blocks gives d_J=1024N).

A *dual-head patch cross-attention projector* — the only trainable module — consists of a shared trunk (linear embedding to width 2048 with LayerNorm, one 8-head cross-attention layer) queried by two sets of learned query tokens:

- **Text head:** N_p queries produce tokens in the backbone's context dimension. They are *appended to the frozen null-prompt embeddings*, so the transformer sees $[\,\mathbf{c}_\varnothing;\ P_{\mathrm{text}}(\mathbf{F})\,]$ in its text stream — the projected features play the role of the prompt.
- **VAE head:** one query per packed-latent position (1024 at 512^2, dim. 128) produces pseudo image-latent tokens that are concatenated to the transformer's latent token sequence with their own positional ids.

Crucially, the VAE encoding of $\mathbf{y}$ itself is *not* concatenated to the latent stream in our main configuration: projected V-JEPA tokens are the only pathway from the input image to the backbone. Velocity predictions are read off the noisy-latent positions only.

3.3. Two-stage training

Stage A — projector alignment. With every network frozen, the projector is trained (20K steps) to map V-JEPA tokens of a source image toward two targets: (i) the text head regresses the frozen Klein text-encoder embeddings of a Qwen3-VL caption of the clean image (MSE, weight 1.0); (ii) the VAE head regresses the packed VAE latents of the clean image (MSE, weight 1.0) plus a VGG perceptual loss on the decoded prediction (weight 0.1). The source image is the clean image by default; *LQ-source* and *SwinIR-pre* variants align from degraded inputs to close the train/test domain gap (Sec. 5.3).

Stage B — flow-matching fine-tuning. The aligned projector is hot-plugged into the frozen backbone and fine-tuned end-to-end with Eq. (1) on blur/sharp pairs (50K steps, AdamW, lr 2×10^{-5}, constant schedule with 500-step warmup, per-GPU batch 2 on 2×A100, bf16, 512^2 crops, gradient clip 1.0, guidance 1.0). Inference runs 50 Euler steps.

3.4. Controls and variants

Encoder swaps. DINOv2-L (518^2) [12] and CLIP ViT-L/14 (224^2) [13] replace V-JEPA behind a single-head (text-stream) projector. VL-JEPA [14][verify open-vljepa citation] contributes a global semantic embedding that *replaces* the null prompt, testing semantic-only conditioning. **Image-conditioned LoRA control.** The no-context control concatenates LQ VAE latents to the latent stream (Kontext-style) and tunes the backbone with LoRA (r=128); it represents the conventional conditioning route. It is not parameter-matched to the projector and is reported as a reference point, not an ablation. **Oracle source.** Extracting V-JEPA features from the ground-truth sharp image (unavailable at inference) upper-bounds what blur-robust features could achieve. **Blur-map adapters.** A related variant (ContextFlux-BlurMap) injects a blur-map estimate and V-JEPA features through Coupler/PGA/PCM adapters without Stage-A alignment; we report it as an architectural alternative.

4. V-JEPA FEATURES UNDER BLUR

We probe whether V-JEPA representations are usable conditioning for deblurring *before* training anything. All probes

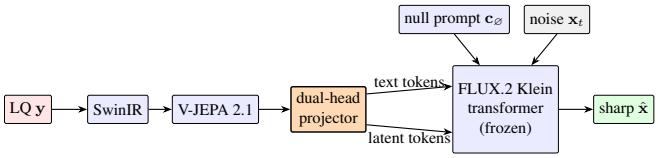


Fig. 1. **ContextFlux (Stage B).** Only the projector (orange) trains; every other module is frozen. The LQ image reaches the backbone *only* through projected V-JEPA tokens: the text head's tokens are appended to the null-prompt text stream, the VAE head's tokens to the noisy-latent token sequence. Stage A pre-aligns the two heads to caption embeddings and clean-image VAE latents. [Draft figure; polish and add a Stage-A panel before submission.]

Table 1. V-JEPA feature stability vs. input resolution (*single* validation image; cosine against the 384px reference).

Res.	Tokens	Grid	Mean $\|f\|_2$	Cos vs. 384
224	196	14^2	58.10	0.840
384	576	24^2	60.37	1.000
512	1024	32^2	60.77	0.910
768	2304	48^2	60.82	0.906

use the frozen `vjepa2_1_vit_large_384` encoder.

Blur invariance. On matched sharp/blurred pairs from our benchmark, the mean per-patch cosine similarity between V-JEPA tokens of the sharp image and its blurred counterpart is 0.724 ± 0.142 and 0.954 ± 0.049 on two validation pairs — i.e. blur-induced drift is real but the representation remains strongly aligned with the sharp image. [These are 2 image pairs. Scale to 100–500 pairs for all three encoders (plan step 7) and replace this paragraph with the resulting figure: cosine-similarity distributions + per-SVD-component energy ratio σ_{lq}/σ_{gt} and subspace alignment for V-JEPA vs. DINOv2 vs. CLIP, from `playground/analyze_jepa_svd.py`. A third legacy probe pair gave an inconsistent near-zero cosine and is excluded pending re-verification.]

Resolution robustness. V-JEPA 2.1 uses RoPE positional encoding, so we can run it above its 384px pre-training resolution. On a single validation image, patch tokens extracted at 224–768px keep a 0.84–0.91 mean cosine similarity to the 384px reference (Table 1), with stable feature norms above 384px. We treat this as license to feed 512^2 crops without resizing, not as evidence of equivalence. [Extend to a dataset average.]

5. EXPERIMENTS

5.1. Setup

Data. All models train and evaluate on our synthetic motion-blur benchmark (Deblur-MIOR): sharp/blurred pairs with spatially varying motion blur, 512^2 training crops. [Fill dataset construction summary + train/val counts; report the full validation split size used for evaluation.] **Metrics.** PSNR, SSIM [19], LPIPS [20], DISTS, and no-reference NIQE/MUSIQ, computed on full validation images with identical protocol (fixed seed, 50 Euler steps) for every method. [All numbers below come from the unified evaluation of plan step 5 — never from training-time validation, which used 5 batches.] **Baselines.** NAFNet [2] and Restormer [1] trained on the same data; a diffusion-prior baseline [pick: in-repo OSEDiff or DiffBIR repro]; and the image-conditioned LoRA control of Sec. 3.4.

Table 2. Deblurring on the Deblur-MIOR validation split. All — cells pending the unified evaluation (checkpoints verified present on scratch, 2026-08-26). †CLIP checkpoint stops at 34K/50K steps.

Method	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	MUSIQ↑
Blurred input	—	—	—	—	—
NAFNet	—	—	—	—	—
Restormer	—	—	—	—	—
LoRA + LQ latents	—	—	—	—	—
ContextFlux-CLIP†	—	—	—	—	—
ContextFlux-DINOv2	—	—	—	—	—
ContextFlux-V-JEPA	—	—	—	—	—

Table 3. Ablations (V-JEPA configuration unless noted). "Source" = image fed to the frozen encoder. GT source is an oracle (unavailable at inference).

Variant	PSNR↑	SSIM↑	LPIPS↓
Feature depth (concat. last N blocks)			
N=1	—	—	—
N=4	—	—	—
Feature source			
LQ (SwinIR-pre)	—	—	—
GT (oracle)	—	—	—
Stage-A alignment source			
GT-aligned	—	—	—
LQ-aligned	—	—	—
Projector heads			
text only	—	—	—
text + VAE	—	—	—

5.2. Main results

Table 2 compares conditioning encoders under an identical frozen backbone and training budget (50K steps) against regression and LoRA-tuned baselines. [Write the comparison narrative after numbers land. If V-JEPA does not decisively beat DINOv2, pivot framing to the controlled-study angle per plan §5; the analysis of Sec. 4 then serves as the predictor of encoder ranking.]

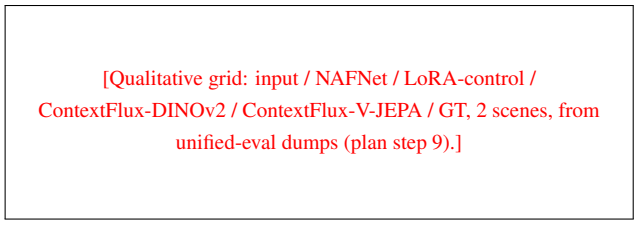


Fig. 2. Qualitative comparison. [pending]

5.3. Ablations

[Run status (2026-08-26): $N{=}1$, $N{=}4$ checkpoints exist; GT-oracle Stage-B and LQ-aligned Stage-B launch per plan §5 day-0/1; heads ablation needs a text-only Stage-B rerun or reuse of an existing run — verify config provenance before filling. VL-JEPA row dropped unless its rerun is funded by the timeline (checkpoint stops at 6K).]

Blur-map adapter variant. The only number currently recoverable from a surviving checkpoint of this family: ContextFlux-BlurMap at 50K steps scores PSNR 24.44 ± 3.62 dB, SSIM 0.740 ± 0.167, LPIPS 0.164 ± 0.091 on its own 5-image validation subset (one low-quality outlier depresses the mean). It is *not* comparable to Table 2 (different split size and protocol) and will be re-evaluated under the unified protocol. [Replace with unified-protocol number; then this caveat shrinks to one clause.]

5.4. Qualitative results

6. CONCLUSION

We presented ContextFlux, which conditions a fully frozen flow-matching transformer for motion deblurring exclusively through self-supervised V-JEPA tokens mapped by a lightweight aligned projector — no captions, no VLM at test time, and in the main configuration no image-latent conditioning. Our feature analysis shows V-JEPA representations degrade gracefully under blur, motivating their use as a restoration conditioning interface, and our controlled encoder comparison quantifies this choice. [Sharpen once results land; state limitations: synthetic-blur scope, sampling cost, single dataset.]

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